



SAPTHAGIRI NPS
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REPORT ON CASE STUDY

“AGILE SCRUM PROCESS”

Submitted in the partial fulfilment for the award of the degree of

BACHELOR OF ENGINEERING

IN

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Submitted by

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DEPARTMENT OF CSE - AIML

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CERTIFICATE

Certified that the MINI PROJECT entitled “**AGILE SCRUM PRECESS**” is carried out by **KISHAN K**, bearing **24SUUBELAML09**, debonair student at **School of Engineering and Technology** in partial full fulfillment for the award of **Bachelor of Engineering** in department of **CSE - AIML** of **SAPTHAGIRI NPS** University, during the academic year **2025-2026**. It is certified that all corrections/suggestions indicated in the Internal Assessment have been incorporated in the report submitted. The mini project report has been approved as it satisfies the academic requirements in respect of mini project prescribed for the Bachelor of Engineering Degree.

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ABSTRACT

"Scrum" is one of the most widely used Agile frameworks for project management and product development. It is based on the principles of transparency, inspection, and adaptation, which help organizations improve efficiency and respond quickly to changing requirements. Scrum divides work into short development cycles called sprints, allowing teams to deliver products incrementally while continuously reviewing and improving their processes.

This report focuses on understanding the Scrum process and its importance in modern project management. It examines the structure of Scrum, including its roles, events, and artifacts, and explains how it helps organizations achieve better collaboration, faster delivery, and improved adaptability. The report also presents a case study of Gillette, a global manufacturing company, to demonstrate how Scrum can be successfully applied beyond software development.

The case study highlights how Gillette used Scrum to simplify complex product development processes, improve cross-functional collaboration, increase transparency, and accelerate innovation. The findings of this report show that Scrum is not limited to software engineering but is also effective in physical product development and manufacturing environments. The study concludes that Scrum is a powerful framework for organizations seeking continuous improvement, operational efficiency, and competitive advantage in today's dynamic business environment.

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Chapter 1: INTRODUCTION

Overview of Scrum

Scrum is one of the most widely used frameworks under Agile methodology for project management and product development. It is designed to help teams work together efficiently, deliver value quickly, and continuously improve through regular feedback.

Scrum is called an empirical process framework because it is based on practical experience, observation, experimentation, and continuous learning. Instead of following a fixed plan from beginning to end, Scrum encourages teams to work in small steps, inspect results frequently, and make necessary improvements.

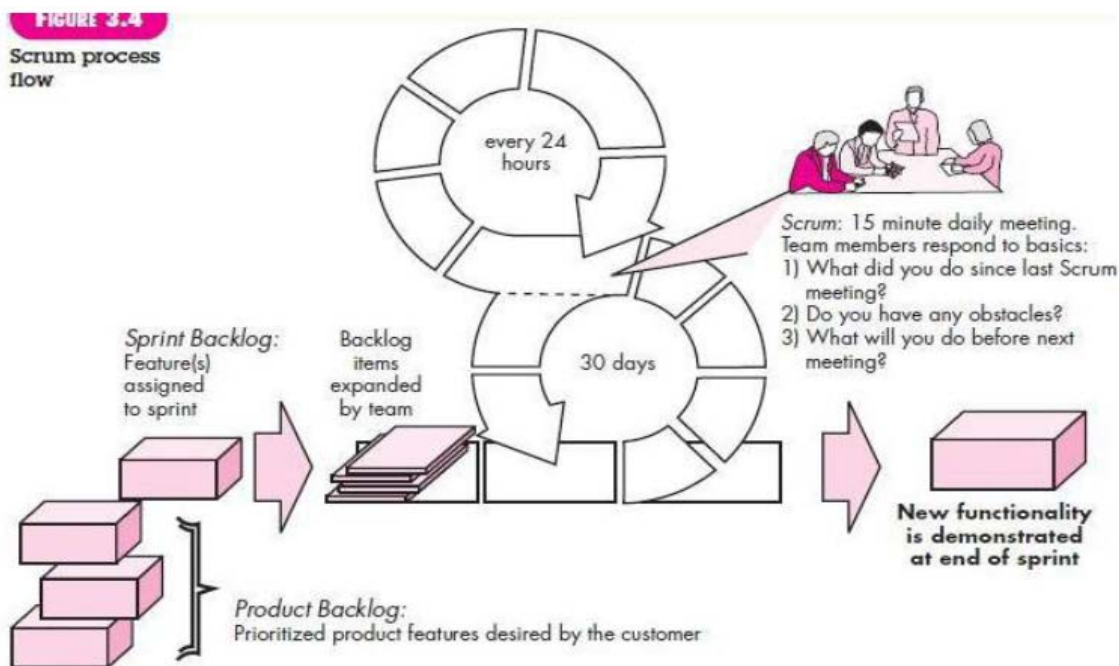
The word “Scrum” is borrowed from the game of rugby, where players come together in a tight formation to move the ball forward. Similarly, in Scrum, team members work closely together toward a shared product goal.

Importance of Scrum

In today’s fast-changing business environment, customer needs and market conditions change quickly. Traditional project management methods often struggle to handle these changes because they follow rigid planning structures.

Scrum provides flexibility by allowing:

This makes Scrum highly suitable for industries that require innovation and adaptability.



CHAPTER 2: BACKGROUND STUDY

Evolution of Project Management

Project management has undergone major changes over the years as organizations have searched for better ways to develop products and complete projects efficiently. In the early stages, most companies followed traditional project management methods such as the Waterfall model. This method follows a fixed sequence where each phase, including planning, designing, development, testing, and deployment, is completed one after another. While this method works well for projects with stable and clearly defined requirements, it becomes less effective when changes are needed during the development process. As industries became more dynamic and customer demands started changing rapidly, traditional methods began showing limitations. Delays, lack of flexibility, and late error detection became common problems. These challenges created the need for more adaptive and iterative project management approaches.

Emergence of Agile

Agile methodology emerged as a solution to the shortcomings of traditional project management approaches. It was introduced to provide flexibility, faster delivery, and continuous improvement. Agile focuses on delivering work in smaller increments rather than waiting until the end of the project to produce final results. This allows teams to receive regular feedback and make necessary improvements throughout the development process. Agile promotes collaboration between team members and customers, ensuring that products align with actual user needs. The Agile Manifesto emphasizes values such as customer collaboration, responding to change, and prioritizing working solutions over excessive documentation. This approach transformed the way organizations manage projects, particularly in industries where adaptability is essential.

Scrum as an Agile Framework

Among various Agile frameworks, Scrum has become the most widely used because of its simplicity and effectiveness. Scrum provides a structured yet flexible framework for managing projects through short development cycles called sprints. Each sprint focuses on delivering a small but valuable part of the final product. This iterative approach allows teams to review progress frequently and adapt quickly to changes. Scrum is built on transparency, inspection, and adaptation, which enable teams to continuously improve their performance. Industry studies indicate that nearly 87 percent of organizations using Agile have adopted Scrum, and around 71 percent of software development teams globally rely on it. The widespread use of Scrum demonstrates its effectiveness in improving project efficiency and product quality.

Scrum Beyond Software Development

Although Scrum was originally developed for software development, its principles can be applied across many industries. Organizations in manufacturing, healthcare, education, marketing, and research have successfully implemented Scrum to improve efficiency and adaptability. The framework's emphasis on collaboration, transparency, and continuous improvement makes it useful in any environment that involves complex problem-solving. Gillette's case study is an excellent example of Scrum being applied in physical product development. This demonstrates that Scrum is not limited to digital products but can also drive innovation in manufacturing and product design.

Why Organizations Adopt Scrum

Organizations adopt Scrum because it provides several advantages over traditional project management methods. One of the primary reasons is faster product delivery. Scrum allows teams to release working product increments frequently instead of waiting until the end of the project. This helps companies bring products to market faster and remain competitive. Another reason is improved customer satisfaction, as frequent feedback ensures that the product meets customer expectations. Scrum also enhances collaboration by encouraging cross-functional teamwork and regular communication. It improves transparency by making progress visible to all stakeholders and reduces risks by identifying problems early. Additionally, Scrum allows organizations to respond effectively to changing market conditions, making it highly suitable for today's fast-paced business environment.

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Chapter 3: PROBLEM DEFINITION AND OBJECTIVES

Problem Definition

Large organizations often face several challenges when managing complex projects. One major issue is process complexity, where multiple stages and excessive procedures slow down execution. In many cases, different departments work independently, leading to communication gaps and lack of coordination. This creates delays and increases the chances of misunderstandings. Decision-making can also become slow because approvals often pass through multiple management levels. Another common problem is the lack of transparency, which makes it difficult for stakeholders to track project progress and identify obstacles. Traditional project management systems are often rigid, making it difficult to adapt to changing customer requirements or market conditions. These challenges reduce efficiency, increase costs, and negatively affect product quality.

Need for Scrum

To overcome these challenges, organizations require a project management framework that is flexible, transparent, and collaborative. Scrum addresses these needs by breaking large projects into smaller, manageable tasks that can be completed within short time periods. This allows teams to identify problems early and make quick adjustments. Scrum encourages continuous communication through regular meetings and ensures that all team members remain aligned with project goals. By promoting transparency and adaptability, Scrum helps organizations simplify workflows and improve overall efficiency. It is particularly valuable in environments where rapid change and innovation are essential.

Objectives of the Study

The primary objective of this study is to understand the Scrum framework and analyze its practical application in real-world organizations. This report aims to examine how Scrum improves project management by enhancing collaboration, increasing transparency, and supporting continuous improvement. Another objective is to analyze Gillette's implementation of Scrum and understand how it addressed organizational challenges. The study also seeks to identify the key benefits achieved through Scrum adoption, such as faster delivery, improved teamwork, and better adaptability. Furthermore, it aims to explore how Scrum can be applied beyond software development in industries such as manufacturing and product innovation.

Scope of the Study

This study focuses on understanding Scrum as an Agile framework and examining its implementation through Gillette's case study. It covers the principles, structure, roles, and events of Scrum, as well as its practical outcomes in a large manufacturing organization. The study provides insights into how Scrum can simplify complex processes and improve efficiency. However, it is limited to conceptual analysis and case study observation rather than direct field research or practical implementation.

Chapter 4: SYSTEM ARCHITECTURE / CONCEPTUAL MODEL

Overview of the Scrum Framework

The Scrum framework provides a structured process for managing projects through iterative cycles. It is designed to ensure regular progress, continuous feedback, and adaptability. Scrum divides work into small development periods called sprints, which usually last two to four weeks. At the end of each sprint, the team delivers a functional product increment. This structure allows teams to monitor progress closely and make improvements continuously. The framework includes defined roles, events, and artifacts that work together to support effective project execution.

Scrum Roles

The Scrum framework includes three important roles: Product Owner, Scrum Master, and Development Team. The Product Owner is responsible for defining project requirements and prioritizing tasks based on customer needs. This role ensures that the team focuses on delivering maximum value. The Scrum Master acts as a facilitator who ensures that Scrum practices are followed correctly. The Scrum Master removes obstacles, supports the team, and helps maintain smooth workflow. The Development Team is responsible for completing tasks and delivering product increments. Team members work collaboratively and are self-organized, meaning they manage their own work to achieve sprint goals.

Scrum Artifacts

Scrum uses specific artifacts to manage and track project work. The Product Backlog is a prioritized list of all tasks, features, and requirements for the project. It serves as the master task list. The Sprint Backlog contains the tasks selected for completion during a specific sprint. It helps the team stay focused on short-term goals. The Product Increment is the completed output delivered at the end of each sprint. These artifacts provide visibility into project progress and help maintain transparency throughout the development process.

Scrum Events

Scrum includes several structured events that ensure regular communication and continuous improvement. Sprint Planning is conducted at the beginning of each sprint to select tasks and define objectives. The Daily Scrum is a short meeting where team members discuss progress, challenges, and next steps. Sprint Review is held at the end of the sprint to demonstrate completed work and gather feedback. Sprint Retrospective allows the team to reflect on their performance and identify areas for improvement. These events create a cycle of planning, execution, evaluation, and adaptation.

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Chapter 5: CASE STUDY – GILLETTE

Introduction to Gillette's Scrum Transformation

Gillette, a globally recognized brand under Procter & Gamble, is known for its innovation in grooming products such as razors, blades, and personal care items. As a large manufacturing organization, Gillette handles highly complex product development processes that involve multiple departments such as research and development, engineering, packaging, marketing, and manufacturing. Managing coordination across these departments often became challenging due to process complexity and delayed communication. To address these challenges, Gillette explored the use of Scrum and Agile principles in its product development operations. This transformation demonstrated that Scrum is not limited to software development and can also be applied effectively in physical product innovation environments. According to the presentation, Gillette used Scrum to improve transparency, simplify processes, and increase innovation speed across its teams .

Challenges Faced by Gillette

Before adopting Scrum, Gillette faced several operational difficulties that affected its ability to innovate quickly. One of the major challenges was the complexity of product development cycles. Since the company develops physical products, every innovation required careful coordination between multiple teams, which often resulted in delays. Departments frequently worked in isolation, creating communication gaps and reducing overall efficiency. Decision-making was also slower because approvals had to pass through multiple levels of management. This lack of transparency made it difficult for teams to identify blockers early and solve issues before they became larger problems. Traditional project management methods added unnecessary layers of bureaucracy, making the entire development process rigid and less adaptable to change. These challenges reduced the company's ability to respond quickly to customer needs and market demands.

Implementation of Scrum at Gillette

To overcome these issues, Gillette introduced Scrum practices into its product development process. The company created cross-functional teams that included members from different departments such as R&D, engineering, packaging, manufacturing, and marketing. These teams worked collaboratively throughout the product development cycle rather than operating separately. Scrum events such as sprint planning, daily stand-up meetings, sprint reviews, and retrospectives were adopted to improve communication and track progress regularly. This iterative way of working enabled teams to focus on delivering smaller outcomes in shorter cycles, making it easier to monitor progress and adapt quickly when changes were needed. Scrum also increased visibility across the organization, allowing everyone involved to clearly understand priorities, challenges, and next steps. Leadership support played an important role in ensuring successful implementation, as executives encouraged experimentation and empowered teams to make faster decisions.

Chapter 6 : PROBLEM CONTEXT

Modern Business Environment

The modern business environment is highly dynamic and competitive, requiring organizations to respond quickly to changing customer needs, technological advancements, and market demands. Companies are under constant pressure to innovate, reduce costs, improve quality, and deliver products faster than their competitors. In such an environment, traditional project management methods often struggle because they are rigid and slow to adapt. Businesses today require systems that allow flexibility, continuous learning, and quick decision-making. This creates the need for frameworks like Scrum, which support adaptability and continuous improvement.

Organizations that fail to respond to rapid market changes risk losing customer trust and competitive advantage. Therefore, project management approaches must enable faster execution, better collaboration, and efficient resource utilization. Scrum provides a solution by allowing teams to work iteratively and make improvements throughout the development process.

Challenges in Traditional Product Development

Traditional product development approaches usually follow a sequential process where one phase is completed before the next begins. Although this structure may seem organized, it often creates delays and limits flexibility. Problems are usually discovered only at later stages, making them expensive and difficult to fix. Communication gaps between departments can further increase inefficiencies.

In large organizations, traditional development processes often involve multiple layers of approval and complex workflows. This slows down decision-making and reduces the ability to respond quickly to changes. Teams may work in isolation, creating departmental silos that prevent effective collaboration. As a result, organizations often experience delays, reduced productivity, and lower innovation potential.

These limitations make traditional approaches unsuitable for industries where speed, adaptability, and innovation are essential.

Need for Agile Transformation

To remain competitive, organizations like Gillette needed to transform their working methods. Agile transformation became necessary to improve flexibility, simplify workflows, and enhance communication. Scrum was introduced as a practical solution because it encourages iterative progress, regular feedback, and team accountability.

The adoption of Scrum allowed Gillette to address organizational complexity and create a more transparent development environment. Agile transformation also helped shift the company's culture toward collaboration and continuous improvement. This change was essential for managing modern product development challenges effectively.

Chapter 7: OUTCOMES AND BENEFITS

Improved Collaboration

One of the most significant outcomes of Scrum implementation is improved collaboration among team members. Scrum encourages cross-functional teamwork, where individuals from different departments work together toward a common objective. This reduces communication gaps and ensures better coordination throughout the project lifecycle.

At Gillette, collaboration improved because teams from research, engineering, manufacturing, and marketing were able to work closely rather than functioning independently. This created a more unified working environment and helped reduce misunderstandings and delays.

Increased Transparency

Transparency is one of the core principles of Scrum. By making tasks, progress, and challenges visible to all stakeholders, Scrum ensures that everyone has a clear understanding of the project's status.

At Gillette, increased transparency allowed teams to identify blockers early and resolve issues before they became major problems. This visibility also improved trust among departments and helped management make better-informed decisions.

Transparency supports accountability and ensures that all team members remain aligned with organizational goals.

Faster Decision-Making

Scrum improves decision-making by creating shorter feedback loops and empowering teams to take ownership of their work. Since progress is reviewed regularly, problems are identified quickly and solutions can be implemented immediately.

Gillette experienced faster decision-making because Scrum reduced unnecessary bureaucracy and encouraged direct communication between teams. This helped accelerate product development and reduced delays caused by slow approval processes.

Faster decision-making is essential for maintaining competitiveness in fast-changing markets.

Improved Innovation

By encouraging experimentation and continuous feedback, Scrum creates an environment that supports innovation. Teams can test ideas quickly, learn from results, and improve continuously.

Gillette benefited from increased innovation because Scrum enabled faster experimentation and better alignment across departments. This helped the company develop products more effectively and remain competitive in the market.

Chapter 8: KEY INSIGHTS AND LEARNINGS

Scrum is a Simplification Tool

One of the most important insights from this study is that Scrum acts as a simplification tool. It helps organizations identify unnecessary complexity and eliminate inefficient processes.

Gillette's experience shows that large organizations often develop complicated systems over time. Scrum helps expose these inefficiencies and encourages teams to focus on value creation.

Simplification improves speed, clarity, and overall productivity.

Scrum is Not Limited to Software Development

Another key learning is that Scrum can be successfully applied beyond software development. Although originally designed for software teams, Scrum's principles are flexible enough to support physical product development, manufacturing, and other industries.

Gillette demonstrated that Agile practices can improve innovation even in environments involving tangible products and long production cycles.

This broad applicability makes Scrum valuable across many sectors.

Leadership Support is Essential

Successful Scrum implementation requires strong support from leadership. Executive sponsorship helps remove organizational barriers, provide resources, and encourage cultural change.

Gillette's transformation succeeded partly because leadership actively supported the Agile adoption process. Without management commitment, teams may struggle to implement Scrum effectively.

This highlights the importance of leadership in driving organizational transformation.

Continuous Improvement Drives Success

Scrum emphasizes regular reflection and improvement through sprint retrospectives. This creates a culture where teams continuously learn from experience and improve their processes.

Gillette's case shows that continuous improvement helps organizations become more efficient over time. Small improvements made consistently can lead to significant long-term success.

This mindset is essential for sustainable growth.

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